UrbanLens — Milestone 5 Engineering Report

Neighborhood Intelligence & Urban Transformation Analysis

Document Status: Final Verified Engineering Report
Milestone: Milestone 5
Git Checkpoint: `a548b6b` (branch `main`)
Project: UrbanLens — Urban Market Intelligence & Housing Affordability Analytics
Date: October 30, 2026

1. Executive Summary

Milestone 5 extends UrbanLens with **hyper-local spatial intelligence and exploratory urban transformation modeling** without disturbing previously established baselines from Milestones 1–4. The module connects real OpenStreetMap (OSM) civic infrastructure features to spatial sub-regions, calculates density indicators, accessibility matrices, and walkability catchments, and establishes an academic indicator registry for urban change analysis.

Summary of Completed Objectives

1. **Interactive Spatial Map**: Fully functional React Leaflet map with approximate bounding-box polygons, real OSM amenity markers, category filtering, and centroid walkability metrics.
2. **Multi-Area Comparison Engine**: Side-by-side comparison interface benchmarking up to 4 areas simultaneously with Recharts visualizations and distance matrices.
3. **Urban Transformation Indicator Registry**: Transparent audit of 12 transformation indicators across 7 domains with strict academic integrity (refusing to claim city-level HPI proves gentrification).
4. **Backend REST APIs**: 7 new Express endpoints providing region metadata, RFC 7946 GeoJSON polygons, detail profiles, comparative benchmarking, and indicator catalogs.
5. **Python Geospatial Analytics**: Pure-math Python pipeline implementing the Haversine formula, bounding-box point-in-polygon assignment, and area density calculations without heavyweight spatial DB dependencies.
6. **Full Regression Verification**:
    - Python unit tests: **116/116 passed** (48 new M5 tests + 68 regression tests)

- Express API tests: **26/26 passed** (6 new M5 tests + 20 regression tests)
- Client production build: **Passed with zero errors** (Vite + TypeScript)

2. Workstream Breakdown & Deliverables

Workstream A: Frontend Applications & Geospatial UI

- **Spatial Neighborhood Intelligence (`client/src/pages/NeighborhoodIntelligence.tsx`)**:
  - Replaced prototype stub with a live React Leaflet map centered on South Mumbai.
  - Interactive polygon layers rendering sub-regions with distinct color palettes and active-selection highlighting.
  - 232 verified OSM amenity nodes with category-specific badges and coordinate popups.
  - Interactive filter bar: All Categories, Schools, Clinics & Hospitals, Transit Stops, and Parking Facilities.
  - Selected Area Deep Dive: Area (km^2 km2), Amenity Density ($features/km^2$ features/km2), 500m Walkshed, and 1km Transit Catchment.
  - Prominent `DataStatusBadge` (`VERIFIED REAL DATA`) and methodology disclosure.
- **Area Comparison Page (`client/src/pages/NeighborhoodComparison.tsx`)**:
  - Multi-select toggle bar allowing users to compare 1 to 4 areas side-by-side.
  - KPI Cards comparing density, feature counts, and walkability.
  - Recharts Bar Charts for amenity density and grouped category breakdowns.
  - Accessibility Distance Matrix displaying Haversine distance in km from each area centroid.
- **Urban Transformation Analysis (`client/src/pages/UrbanTransformation.tsx`)**:
  - Macroeconomic context card displaying official RBI HPI Mumbai metrics (15-year cumulative change, 5-year change, CAGR).
  - Indicator Registry table with status filter (`Verified Real` vs `Data Unavailable`) and category filter.
  - Prominent academic integrity notice explaining why UrbanLens does not assign automated gentrification scores.
- **App Layout & Routing (`client/src/App.tsx`, `client/src/components/layout/Sidebar.tsx`)**:

- Integrated routes: `/neighborhood`, `/neighborhood/compare`, and `/transformation`.
- Added navigation items with Lucide icons (`MapPin`, `Scale`, `Sparkles`).

Workstream B: Express Backend Services & REST APIs

- **API Types (`server/src/types/api.ts`)**:
  - Defined strict TypeScript contracts for `NeighborhoodRegion`, `RegionBbox`, `RegionCentroid`, `NeighborhoodAnalyticsPayload`, `TransformationIndicator`, and `HpiCityTrend`.
- **Neighborhood Service (`server/src/services/neighborhoodService.ts`)**:
  - In-memory cached reading of pre-computed `neighborhood_analytics.json` and `transformation_indicators.json`.
  - Input validation guarding against path traversal or malicious identifiers.
- **Neighborhood Routes (`server/src/routes/neighborhoods.ts`)**:
  - `GET /api/neighborhoods`: Region summaries and metadata.
  - `GET /api/neighborhoods/boundaries`: GeoJSON Polygon FeatureCollection.
  - `GET /api/neighborhoods/:id`: Individual region profile and metrics.
  - `GET /api/neighborhoods/:id/analytics`: Deep-dive analytics with data integrity context.
  - `GET /api/neighborhoods/compare?ids=a,b`: Multi-area comparative query.
  - `GET /api/neighborhoods/amenities`: Preserved M4 baseline raw amenity query.
- **Transformation Routes (`server/src/routes/transformation.ts`)**:
  - `GET /api/transformation/indicators`: Filterable catalog of transformation indicators.
  - `GET /api/transformation/summary`: Aggregate macroeconomic summary and integrity mandate.

Workstream C: Python Geospatial & Transformation Pipelines

- **Geospatial Pipeline (`server/python/geo_pipeline.py`)**:
  - Built using standard Python library (`math`, `csv`, `json`) without requiring heavy C-extensions like GDAL/GEOS or GeoPandas.
  - Great-circle Haversine formula calculation:
    $$d = 2R \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{1-a})$$
    d = 2R · atan2(a

, 1 – a

)

- Equirectangular bounding-box surface area estimation for local coordinates.
- 100% assignment rate: all 232 South Mumbai amenities mapped into 4 sub-regions.
- Outputs `server/python/data/processed/neighborhood_analytics.json`.

- **Transformation Registry Pipeline (`server/python/transformation_indicators.py`)**:
  - Ingests real historical RBI HPI for Mumbai (60 quarters, 2010 to 2025).
  - Computes 15-year cumulative change (+370.39%), 5-year change (+45.45%), and CAGR (10.87%).
  - Catalogs 8 transparently unavailable indicators with documented justifications and potential proxies.
  - Outputs `server/python/data/processed/transformation_indicators.json`.

3. Data Integrity & Provenance Mandate

Domain	Dataset / Indicator	Status	Geographic Level	Source & Provenance
Macro Housing Price	RBI House Price Index	**Real**	City (Mumbai)	Reserve Bank of India DBIE portal (2010–2025)
Local Infrastructure	Civic Amenity Features (232 nodes)	**Real**	Neighborhood Extent	OpenStreetMap via Overpass API (Oct 2026)
Sub-Region Boundaries	Approximate Research Extents	**Documented**	South Mumbai Sub-Regions	Bounding boxes derived from landmark references
Micro Housing Price	Ward-Level Property Prices	**Unavailable**	Ward / Sub-District	Not published in official free Indian statistical releases
Demographics	Ward Population / Income Shifts	**Unavailable**	Ward	Census 2021 unreleased; PLFS is state/district only
Urban Development	Construction Permits / SRA Footprint	**Unavailable**	Ward	MCGM / SRA lack open programmatic machine-readable APIs

> **Ethical Principle Followed**: UrbanLens strictly avoids generating fake ward polygons, fabricating localized price indexes, or labeling neighborhoods as "gentrifying" based on partial data.

4. Verification and Test Results

4.1 Python Unit Tests

Ran test suite covering Haversine distance, bounding box areas, point-in-bbox logic, amenity assignment, nearest facility distances, radius counts, and transformation indicators:

```
Ran 116 tests in 0.423s
OK (116/116 passed)
```

4.2 Express API Integration Tests

Ran 26 automated integration tests validating all endpoints, parameter validation, path traversal prevention, and data payloads:

```
▶ UrbanLens Express API Endpoints
  ✔ 1. GET /api/health → 200, status: ok
  ...
  ✔ 20. GET /api/data/series?geography=Mumbai&metric=hpi_assessment → returns real
observations
  ✔ 21. GET /api/neighborhoods → 200, array of regions with analytics
  ✔ 22. GET /api/neighborhoods/boundaries → 200, GeoJSON FeatureCollection
  ✔ 23. GET /api/neighborhoods/south-mumbai-colaba/analytics → 200
  ✔ 24. GET /api/neighborhoods/compare?ids=south-mumbai-colaba,south-mumbai-fort →
200
  ✔ 25. GET /api/transformation/indicators → 200, array with defined fields
  ✔ 26. GET /api/transformation/summary → 200
✔ UrbanLens Express API Endpoints (184.91ms)
ℹ tests 26 | pass 26 | fail 0
```

4.3 Client Production Build

```
vite v6.4.5 building for production...
✓ 2270 modules transformed.
dist/index.html                   1.01 kB │ gzip:   0.56 kB
dist/assets/index-CXQGMd2l.css   57.84 kB │ gzip:  14.35 kB
dist/assets/index-gKiDNe7U.js   936.82 kB │ gzip: 260.54 kB
✓ built in 17.27s (zero errors)
```

5. Documentation Deliverables

The following engineering documents were created under `docs/`:

- `docs/NEIGHBORHOOD_INTELLIGENCE.md`: Architecture, mathematical formulations, and endpoint specifications.
- `docs/GEOJSON_BOUNDARIES.md`: Detailed bounding-box definitions, coordinates, and MCGM boundary analysis.
- `docs/AMENITY_ANALYTICS.md`: OpenStreetMap amenity taxonomy, counts, and accessibility indicators.

- `docs/URBAN_TRANSFORMATION_METHODOLOGY.md` : Academic rationale, indicator registry breakdown, and anti-fabrication guidelines.

6. Git Commit Checkpoint

- **Commit Hash**: `a548b6b`
- **Commit Message**: `feat: milestone 5 — neighborhood intelligence, geospatial analytics, urban transformation`
- **Files Modified / Created**: 24 files (+3,896 lines, -265 lines)